15/03

FILE HANDLING

What is File Handling?

Accessing the data of the file through program for reading or writing purpose.

How to do we access the files in cprogram?

We have predefined functions to accesses the file in the program.

File Predefined Functions

1.fopen
2.fclose
3.fgetc
4.fgets
5.fread
6.fscanf
7.fputc
8.fputs
9.fwrite
10.fprintf
11. rewind
12.fseek
13.ftell
14.feof

i/p related functions

o/p related function

random accessing funs

```
FILE *stream fopen(const char *,const char *);
```

filename

mode

types of modes

"r" — read only
"w" — write only
"a" — append
"r+" — read + write, append
"w+" — write + read, append
"a+" → append + read, write

} text file

"rb"
"wb"
"ab"
"rb+"
"wb+"
"ab+"

} binary file

"r" --> opened only for reading purpose , we can only open if file existing
if file not existing it will not create a new file.

"w" ---> opened for writing purpose, it opens the file as a newfile
if already file is existing and contains some data , it deletes and opens as new file

"a" ---> opens the file for writing purpose without effecting existing data

RAM
random acess memory

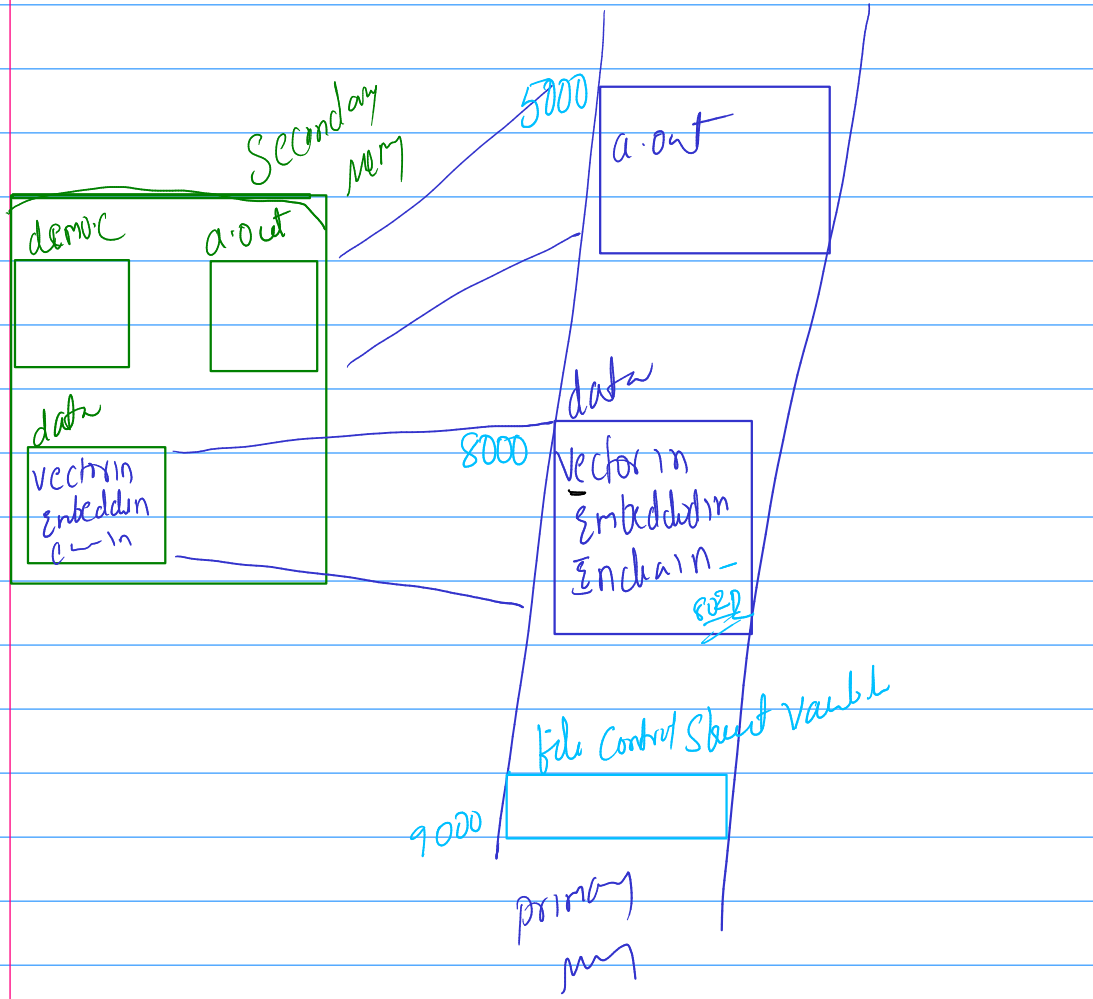
ROM
read only memory

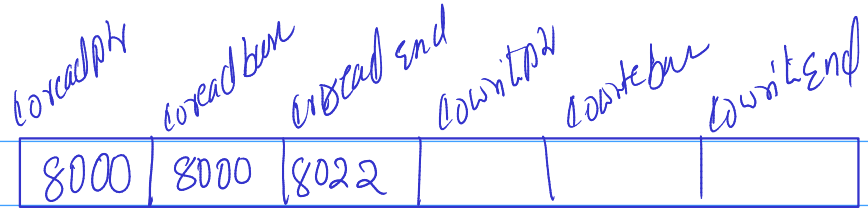


① data seg
② Stack
③ text/code
④ heap

primary memory
volatile

secondary mem
nonvolatile





ch = fgetc(fp);

FILE is an opaque datatype, which is typedef of file control structure.
File control structure implementation varies from one os to another os, so we use file predefined functions

fopen function searches for respective file in the current director or in given path,
if found loads that file in primary memory and creates a file control structure variable to it and returns its address, which we are colecting into FILE *pointer.

else if file not found , it returns NULL

fclose function saves the updation of the file into secondary memory